

# CBSE Class 12 Physical Education — Key Formulas & Facts Sheet

Here is the comprehensive chapter-wise key formulas and facts sheet, highlighting all the crucial numeric values, calculable items, and essential concepts required for the CBSE Class 12 Physical Education board exam.

## Unit 1: Management of Sporting Events

- **Total Matches in Knock-out Tournament:**  $N - 1$  (where  $N$  is the total number of teams) [1].
- **Total Matches in Single League Tournament:**  $\frac{N(N-1)}{2}$  [2].
- **Total Matches in Double League Tournament:**  $N(N - 1)$  [3].
- **Calculation of Byes:** Next highest power of 2 –  $N$  (e.g., if  $N = 11$ , the next power of 2 is 16. Byes =  $16 - 11 = 5$ ).
- **Number of Byes in Upper Half (if  $N$  is odd):**  $\frac{N_b-1}{2}$  (where  $N_b$  is the total number of byes).
- **Number of Byes in Lower Half (if  $N$  is odd):**  $\frac{N_b+1}{2}$ .
- **Number of Rounds in Knock-out:** Equal to the power of 2 used to calculate byes (e.g., for 16 teams,  $16 = 2^4$ , so there are 4 rounds).
- **Number of Rounds in League (Cyclic Method):**
  - If  $N$  is even:  $N - 1$
  - If  $N$  is odd:  $N$

## Unit 2: Children and Women in Sports

- **Female Athlete Triad:** A syndrome consisting of three interrelated conditions: **Osteoporosis** (low bone density), **Amenorrhea** (absence of menstruation), and **Eating Disorders** (like Anorexia or Bulimia) [4].
- **Menarche:** The onset or beginning of the menstrual period in women [5].

## Unit 3: Yoga as Preventive Measure for Lifestyle Disease

- **Obesity:** Prevented/cured by Vajrasana, Hastasana, Trikonasana, Ardhamatseyendrasana [4].
- **Hypertension:** Controlled using relaxation asanas like Shavasana, Makarasana, and Bhujangasana [6].

- **Asthma:** Prevented by asanas that expand the chest, such as Gomukhasana, Chakrasana, and Matsyasana [4].

## Unit 4: Physical Education & Sports for CWSN (Divyang)

- **Special Olympics:** A global movement for children and adults with intellectual disabilities. The **minimum age to participate is 8 years** [7].
- **Deaflympics:** First organized in the year **1924** [5].

## Unit 5: Sports & Nutrition

- **Body Mass Index (BMI) Formula:**  $\frac{\text{Weight (in kg)}}{\text{Height}^2 \text{ (in m}^2\text{)}} [8].$
- **BMI Categories (WHO norms):**
  - Underweight: **< 18.5** [9]
  - Normal weight: **18.5 – 24.9** [9]
  - Overweight: **25.0 – 29.9** [9]
  - Obese: **> 30.0** [9]
- **Basal Metabolic Rate (BMR):** The minimum number of calories needed to maintain basic vital body functions (like breathing, cell production, and blood circulation) when the body is completely at rest [10].
- **Macro & Micro Nutrients:** Proteins repair body cells (building blocks), Carbohydrates are the main source of energy, and Fats are a concentrated source of energy that transport fat-soluble vitamins (A, D, E, K) [11]. Roughage (fiber) provides **0** calories but aids digestion [11].

## Unit 6: Test and Measurement in Sports

- **Harvard Step Test Fitness Index (Long Formula):**  $\frac{100 \times \text{Test duration in seconds}}{2 \times \text{Sum of 3 heart beats in recovery periods}}$  [12]. (*Heartbeats are counted at 1-1.5, 2-2.5, and 3-3.5 minutes after the exercise*).
- **Harvard Step Test Fitness Index (Short Formula):**  $\frac{100 \times \text{Test duration in seconds}}{5.5 \times \text{Pulse count of 1-1.5 Min after Exercise}}$  [13].
- **SAI Khelo India Fitness Test (Age 5-8 years / Class 1-3):** Measures BMI, Static Balance (Flamingo Balance Test), and Coordination (Plate Tapping Test) [14].
- **SAI Khelo India Fitness Test (Age 9-18 years / Class 4-12):** Measures BMI, Speed (50m Dash), Endurance (600m Run/Walk), Flexibility (Sit & Reach), and Muscular Endurance (Partial Curl-Up for 30 seconds [15], Push-Ups for boys, Modified Push-Ups for girls) [14].
- **Rikli & Jones Senior Citizen Fitness Test:**

- **Arm Curl Test Weight:** 8 lbs (pounds) for men, and 5 lbs (pounds) for women [16].
- Other tests include: Chair Stand (lower body strength), Chair Sit & Reach (lower body flexibility), Back Scratch (upper body flexibility), 8-Foot Up & Go (agility), and 6-Minute Walk (aerobic endurance) [13].

## Unit 7: Physiology & Injuries in Sport

- **Normal Resting Heart Rate:** The average healthy adult resting heart rate is **72 beats per minute** [17].
- **Stroke Volume:** The volume of blood pumped out by the heart (left ventricle) in a single beat/stroke [18].
- **Cardiac Output:** The total amount of blood pumped by the heart in one minute. Formula: **Cardiac Output = Stroke Volume × Heart Rate.**
- **Muscle Fibers:**
  - *Slow-Twitch (Type I / Red Fibers):* High resistance to fatigue; best for endurance events (e.g., marathons) [19].
  - *Fast-Twitch (Type II / White Fibers):* Contract rapidly and forcefully; best for speed and explosive strength events (e.g., 100m sprint) [20].

## Unit 8: Biomechanics and Sports

- **Projectile Motion:** The optimal angle of release to achieve maximum horizontal distance is **45°**.
- **Newton's Laws of Motion:**
  1. *Law of Inertia:* An object remains at rest or in motion unless acted upon by an external force [21].
  2. *Law of Acceleration (Momentum):* The rate of change of momentum is proportional to the applied force [21].
  3. *Law of Reaction:* For every action, there is an equal and opposite reaction (e.g., a swimmer pushing water backwards) [22].
- **Levers in Sports:**
  - *Class I Lever:* Fulcrum is in the middle (e.g., neck joint, seesaw).
  - *Class II Lever:* Load/Resistance is in the middle (e.g., calf raise, push-ups).
  - *Class III Lever:* Effort/Force is in the middle (e.g., bicep curl). *Class III is the most common lever in the human body* [23].

## Unit 9: Psychology and Sports

- **Big Five Personality Theory (OCEAN):** Categorizes personality into Openness, Conscientiousness, Extroversion, Agreeableness, and Neuroticism [24].
- **Jung's Classification:** Divides personality into Introverts (reserved, self-aware), Extroverts (sociable, expressive), and Ambiverts [25].
- **Aggression Types:**
  - *Hostile Aggression:* The primary goal is to inflict physical injury or psychological harm on the opponent [26].
  - *Instrumental Aggression:* The goal is to win a reward or the game; harming the opponent is accidental or a byproduct of the effort [26].

## Unit 10: Training in Sports

- **Training Cycles:**
  - **Microcycle:** Short-term training plan, usually lasting **3 to 10 days** [1].
  - **Mesocycle:** Medium-term training plan, generally lasting **3 to 6 weeks** [1].
  - **Macrocycle:** Long-term training plan, lasting from **3 to 12 months** [27].
- **Fartlek Training:** A Swedish word meaning "Speed Play." It is a blend of continuous and interval training used to develop endurance without a strict rest/work ratio [19].

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CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.